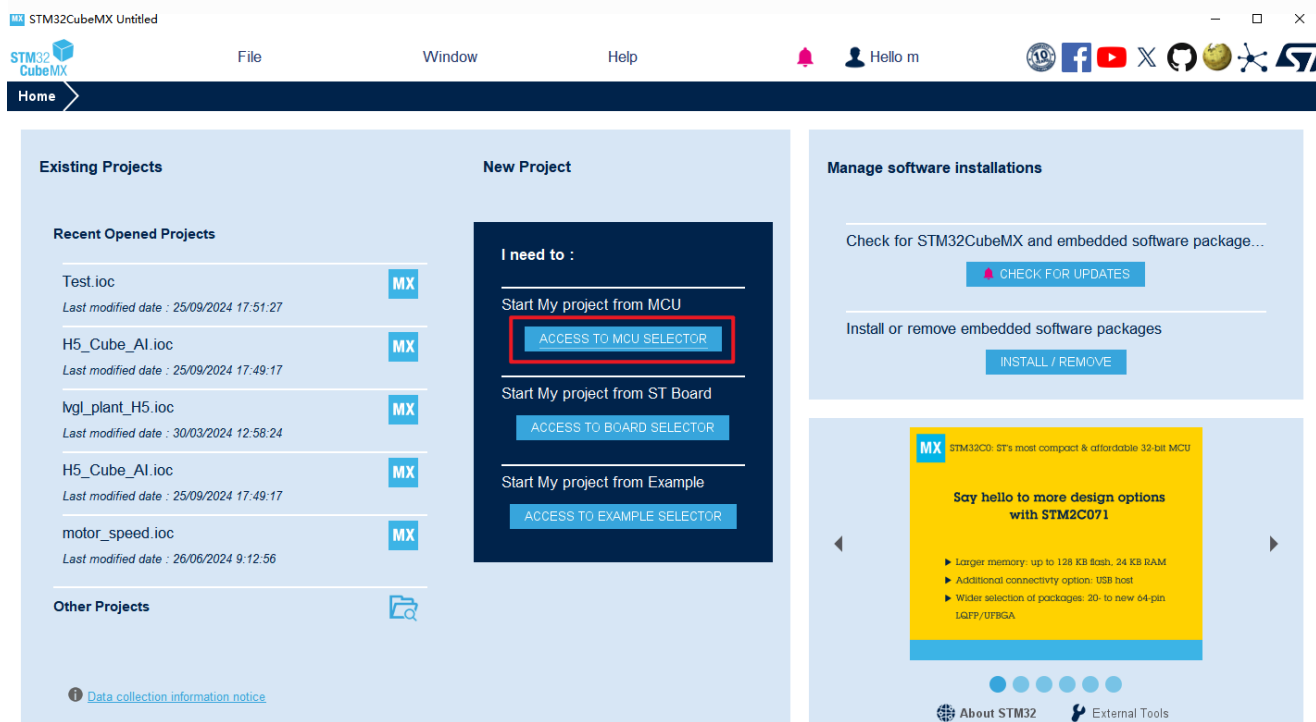


创建第一个工程（Blink）

课程工程案例名称：1_5_DshanMCU_H7R_Blink

使用cubeMX新建工程

双击打开CubeMX，点击 ACCESS TO MCU SELECTOR 输入 STM32H7R7L8H6 选择对应芯片，点击 **Next>** 如下图所示：



MCU/MPU Selector Board Selector Example Selector Cross Selector

MCU/MPU Filters

Commercial Part Number: STM32H7R7L8H6

1. 输入芯片信息

PRODUCT INFO

Segment >

Series >

Line >

Marketing Status >

Price >

Package >

Core >

Coprocessor >

MEMORY

Flash = 64 (kBytes)

EEPROM = 0 (Bytes)

RAM Total = 620 (kBytes)

RAM = 620 (kBytes)

CCM RAM = 0 (kBytes)

STM32H7 Series

STM32H7R7L8H6

High-Performance Arm Cortex-M7 MCU, 600MHz, 64KB Bootflash, 620KB SRAM, with DSP, cache, USB HS PHY, NeoChrom GPU, TFT-LCD

Unit Price for 10KU (US\$): 5.6597

TFBGA 225 13x13x1.2 P 0.8 mm

STM32H7Rxx8 devices are based on the high-performance Arm® Cortex®-M7 32-bit RISC core operating at up to 600 MHz. The Cortex-M7 core features a floating point unit (FPU) which supports Arm double-precision (IEEE 754 compliant) and single-precision data-processing instructions and data types. The Cortex-M7 core includes 32 Kbytes of instruction cache and 32 Kbytes of data cache. STM32H7Rxx8 devices support a full set of DSP instructions and a memory protection unit (MPU) to enhance application security. STM32H7Rxx8 devices incorporate high-speed embedded memories, 64 Kbytes of user flash memory and 128 Kbytes of system flash memory and up to 620 Kbytes of RAM (including 128 Kbytes that can be shared between ITCM and AXI, including 64 Kbytes exclusively ITCM, including 128 Kbyte DTCM, including 64 Kbytes exclusively DTCM, including 32 Kbytes AHB and 4 Kbytes of backup RAM), as well as an extensive range of enhanced I/Os and peripherals connected to APB buses, AHB buses, 2x32-bit multi-AHB bus matrix and a multi-layer AXI interconnect supporting internal and external memory access. To improve application robustness, all

MCUs/MPUs List: 2 items

2. 选择对应芯片

	Commercial Part No	P...	Reference	Marketing...	Unit Price ...	Board	Package	Flash	RAM	IO	Frequency
★	STM32H7R7L8H6		STM32H7...	Active	5.6597		TFBGA 22...	64 kBytes	620 kBytes	152	600 MHz
☆	STM32H7R7L8H6H		STM32H7...	Active	5.8728		TFBGA 22...	64 kBytes	620 kBytes	150	600 MHz

3. 点击"Next>"打开

< Back Next > Finish Cancel

第一次创建需要安装一些软件包，点击我已阅读并同意协议许可，点击“Finish”即安装完成：

MX Licensing Agreement

CubeFw H7RS 1.1.0 License Agreement

Please read and accept the following agreement carefully to finish the installation:

Click here to open the license agreement

☒ I have read, and I agree to the terms of this license agreement

☐ I do not accept the terms of this license agreement

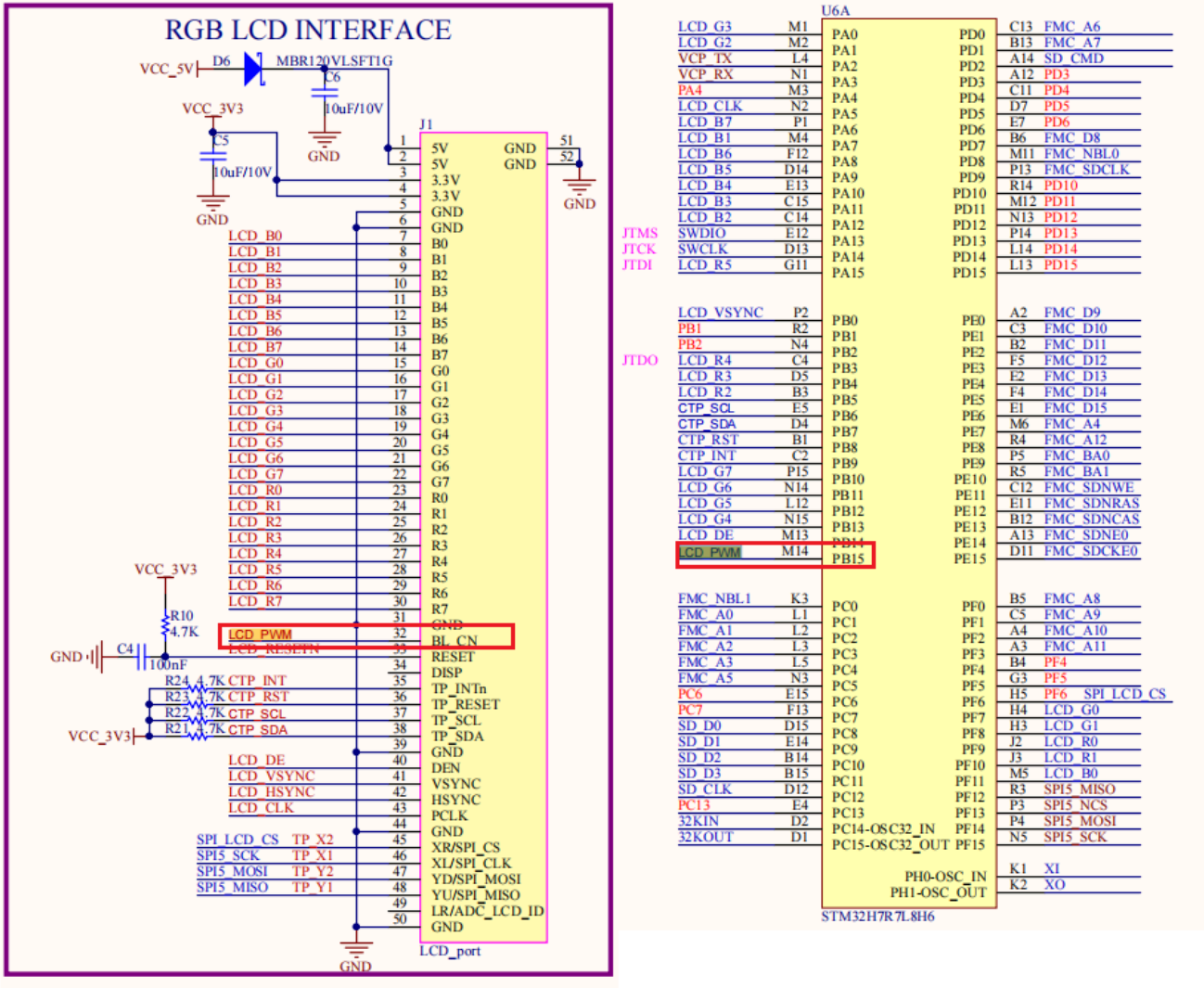
Finish Cancel

安装完成进入工程。

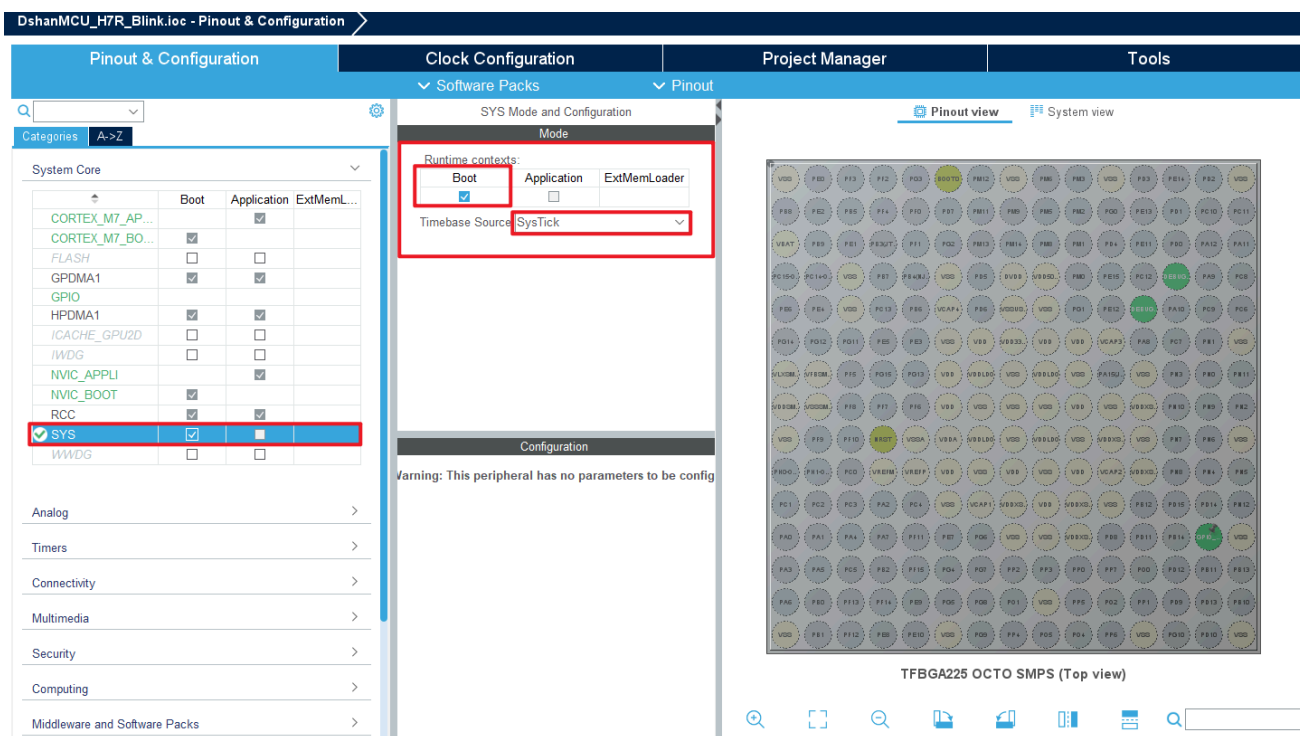
配置工程

本核心板没有板载用户LED灯，所以此工程我们点亮屏幕背光来充当LED灯使用，以便观察现象,没有购买屏幕的用户，能够成功编译烧录即可；

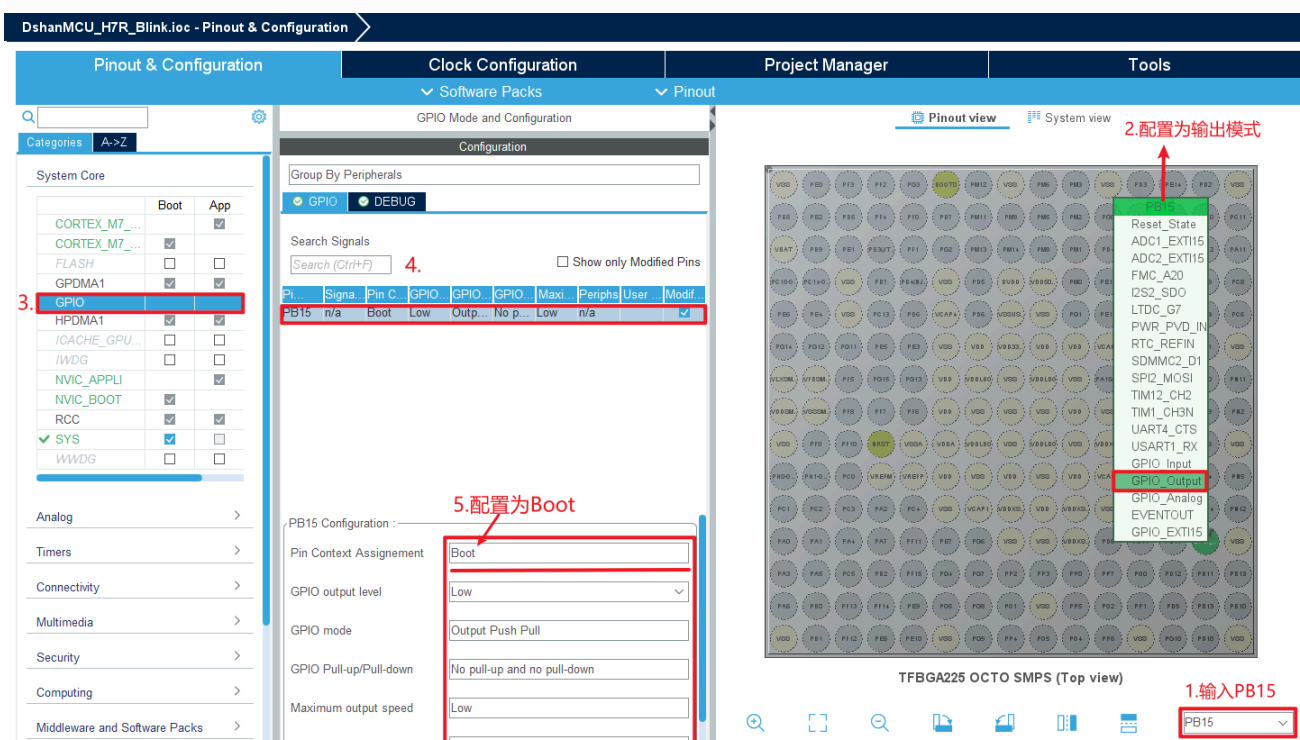
打开“DshanMCU_H7R资料包”找到“05 硬件资料->01核心板原理图”查看屏幕的背光引脚：



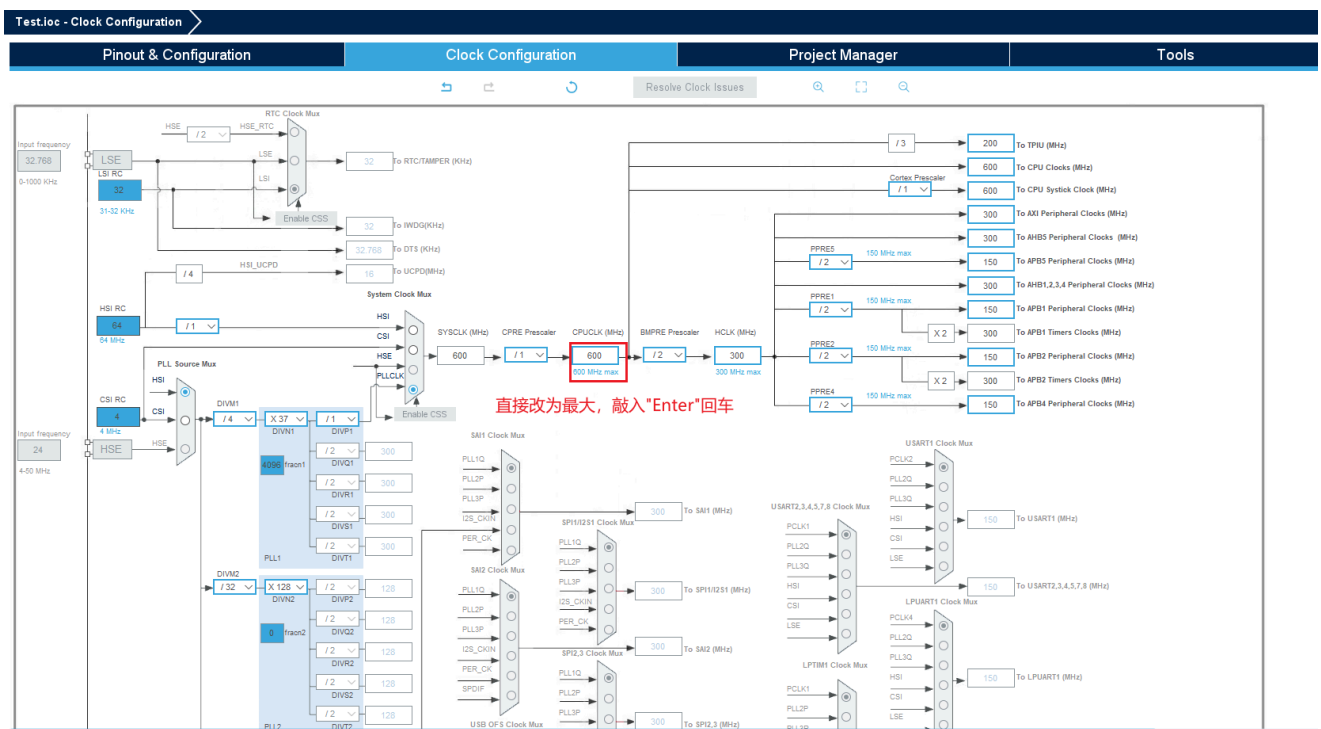
点击SYS选择Boot，时钟源选择SysTick：



这个程序简单控制屏幕背光的亮灭，我们只需要配置背光的引脚，通过原理图我们可以知道屏幕的背光引脚为“LCD_PWM->PB15”，通过搜索框搜索PB15来定位引脚位置，并将PB15配置为“GPIO_Output”输出引脚：



配置时钟：



为了防止出现，烧录以后仿真器无法连接的情况，将Trace and Debug里的 Debug 配置为Serial Wire：

Pin N.	Signal on	Pin Cont.	GPIO out	GPIO mode	GPIO Pul	Maximu
PA13/JT...	DEBUG...	n/a	n/a	n/a	n/a	n/a
PA14/JT...	DEBUG...	n/a	n/a	n/a	n/a	n/a

打开 Project Manager -> Project ,配置文件名，工程属性，存放路径，堆栈设置：

DshanMCU_H7R_Blink.ioc - Project Manager

Pinout & Configuration	Clock Configuration	Project Manager	Tools												
Project	Project Settings Project Name: 1. DshanMCU_H7R_Blink Project Location: D:\Src\DshanMCU_H7R Browse Project Structure: 2. ☒ Boot Project ☒ Appli Project ☐ ExtMemLoader Project Application Structure: Advanced ☐ Do not generate the main() Toolchain Folder Location: 3. D:\Src\DshanMCU_H7R\DshanMCU_H7R_Blink\ Browse Toolchain / IDE: STM32CubeIDE ☒ Generate Under Root														
Code Generator	Linker Settings <table border="1"> <thead> <tr> <th></th> <th>Boot</th> <th>App</th> <th>ExtMemLoader</th> </tr> </thead> <tbody> <tr> <td>Minimum Heap Size</td> <td>4. 0x200</td> <td>0x200</td> <td>0x200</td> </tr> <tr> <td>Minimum Stack Size</td> <td>0x400</td> <td>0x400</td> <td>0x400</td> </tr> </tbody> </table> 2和4跟据需求更改				Boot	App	ExtMemLoader	Minimum Heap Size	4. 0x200	0x200	0x200	Minimum Stack Size	0x400	0x400	0x400
	Boot	App	ExtMemLoader												
Minimum Heap Size	4. 0x200	0x200	0x200												
Minimum Stack Size	0x400	0x400	0x400												
Advanced Settings	Thread-safe Settings Boot ☐ Enable multi-threaded support Thread-safe Locking Strategy: Default - Mapping suitable strategy depending on RTOS selection. Appli ☐ Enable multi-threaded support Thread-safe Locking Strategy: Default - Mapping suitable strategy depending on RTOS selection.														
Boot Path and Debug Authentication	Mcu and Firmware Package Mcu Reference: STM32H7R7L8Hx Firmware Package Name and Version: STM32Cube FW_H7RS V1.1.0														

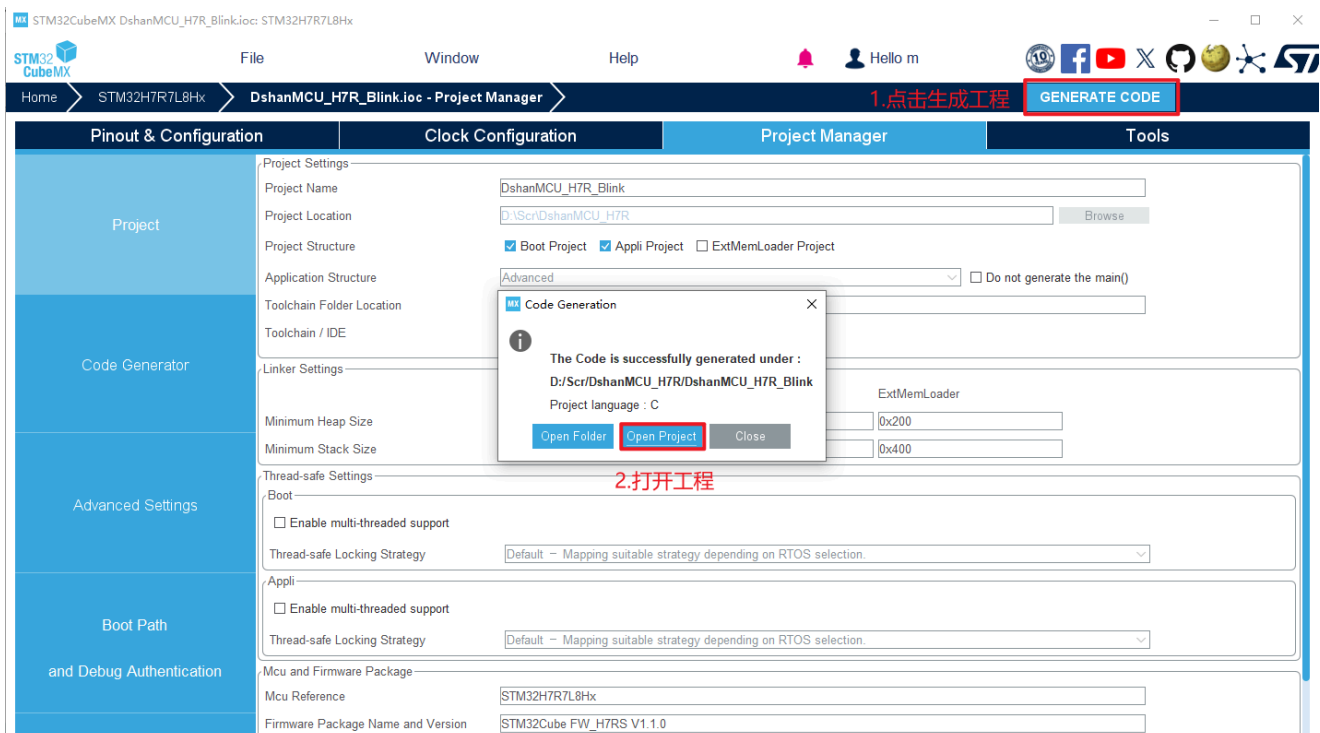
1和3均在我们新建工程时配置好了

打开 **Code Generator** ,选择生成 **.c/.h** 文件，其他默认即可：

TestIoc - Project Manager

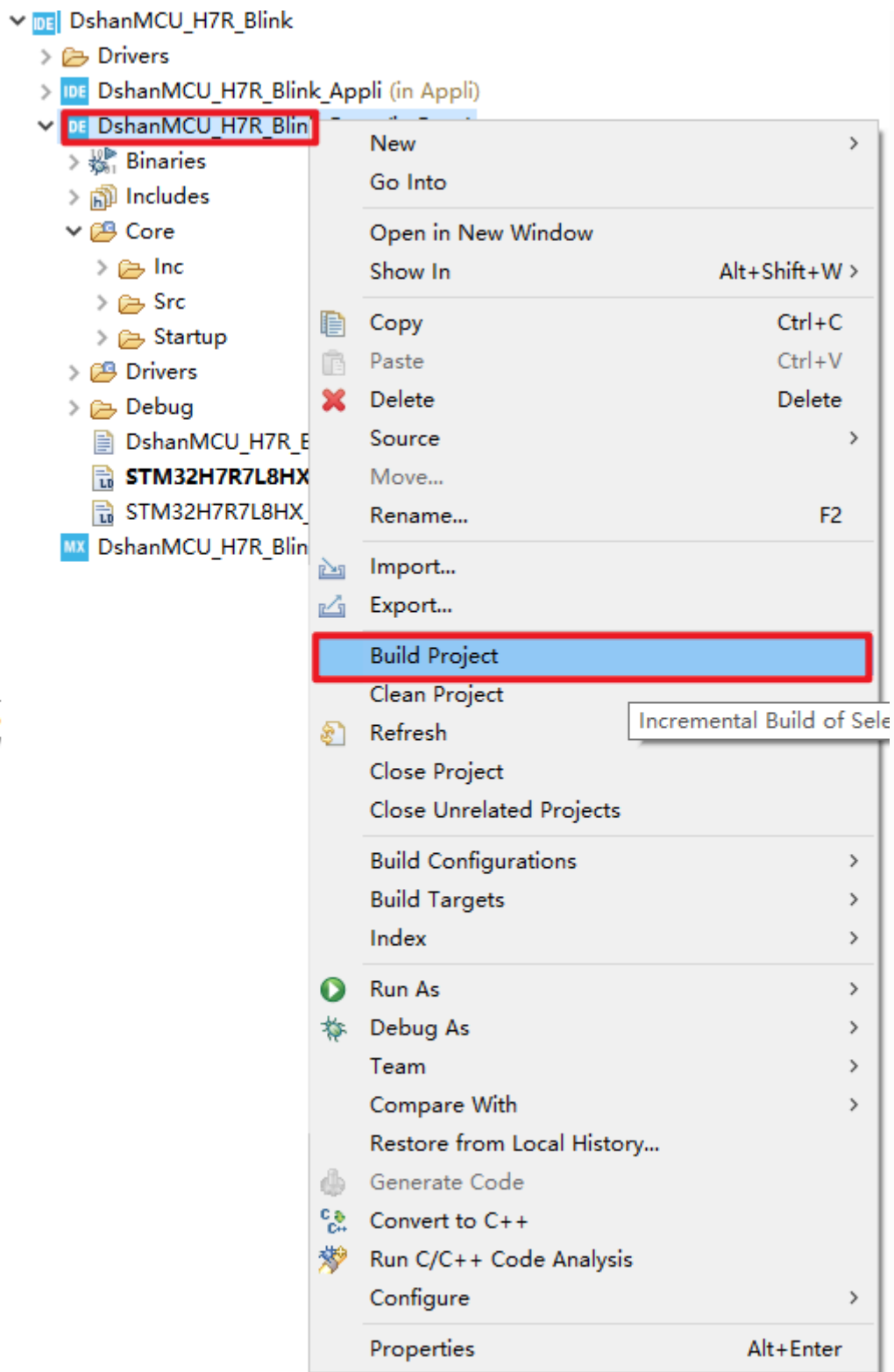
Pinout & Configuration	Clock Configuration	Project Manager	Tools
Project	STM32Cube MCU packages and embedded software packs ☐ Copy all used libraries into the project folder ☒ Copy only the necessary library files ☐ Add necessary library files as reference in the toolchain project configuration file		
Code Generator	Generated files ☒ Generate peripheral initialization as a pair of '.c/.h' files per peripheral 生成.c/.h文件 ☐ Backup previously generated files when re-generating ☒ Keep User Code when re-generating ☒ Delete previously generated files when not re-generated		
Advanced Settings	HAL Settings ☐ Set all free pins as analog (to optimize the power consumption) ☐ Enable Full Assert		
Boot Path and Debug Authentication	User Actions Before Code Generation: Browse After Code Generation: Browse Template Settings Select a template to generate customized code: Settings...		

点击右上角保存并生成工程，会有1个弹窗提示询问我们，是否要打开工程，点击“打开工程”即可，如下图所示：



编译

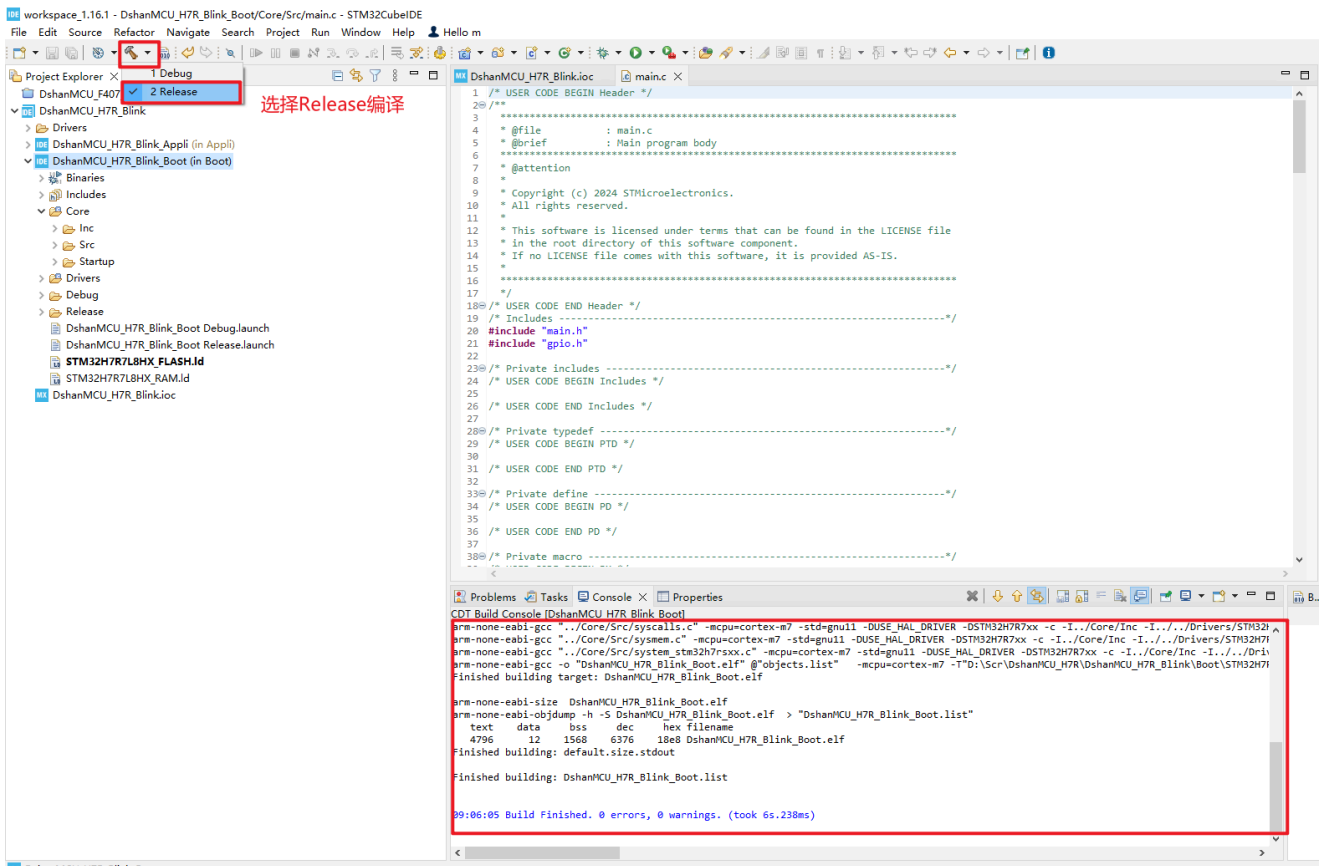
在编译工程之前，我们要先刷新工程文件，避免我们配置没有加载出来，点击Boot程序右键，点击刷新工程，如下图所示：



点击**Boot工程**，打开Src目录下的mian.c，在while循环中添加如下函数：

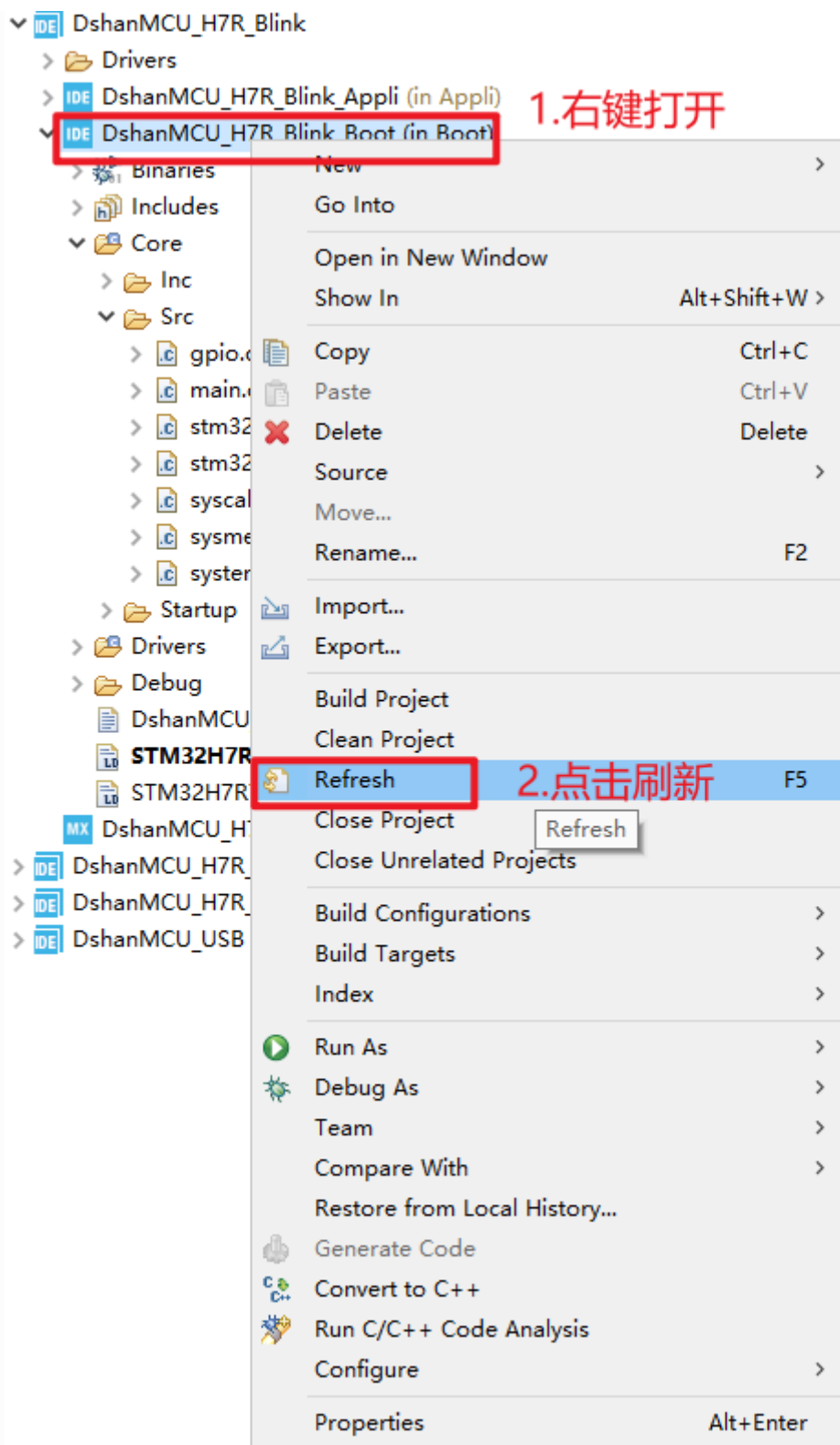
```
HAL_GPIO_WritePin(GPIOB, GPIO_PIN_15, GPIO_PIN_SET);
HAL_Dealy(500);
HAL_GPIO_WritePin(GPIOB, GPIO_PIN_15, GPIO_PIN_RESET);
HAL_Dealy(500);
```

点击工具栏中的编译下来按钮，选择 **ReLaeas** 编译构建程序：



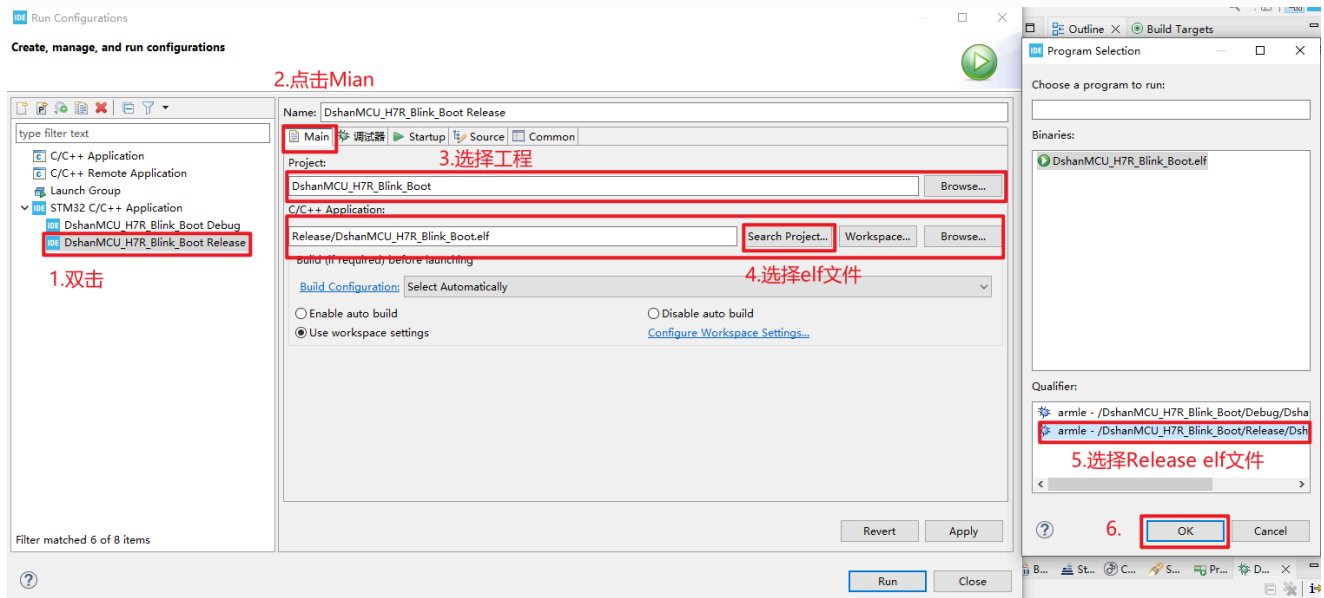
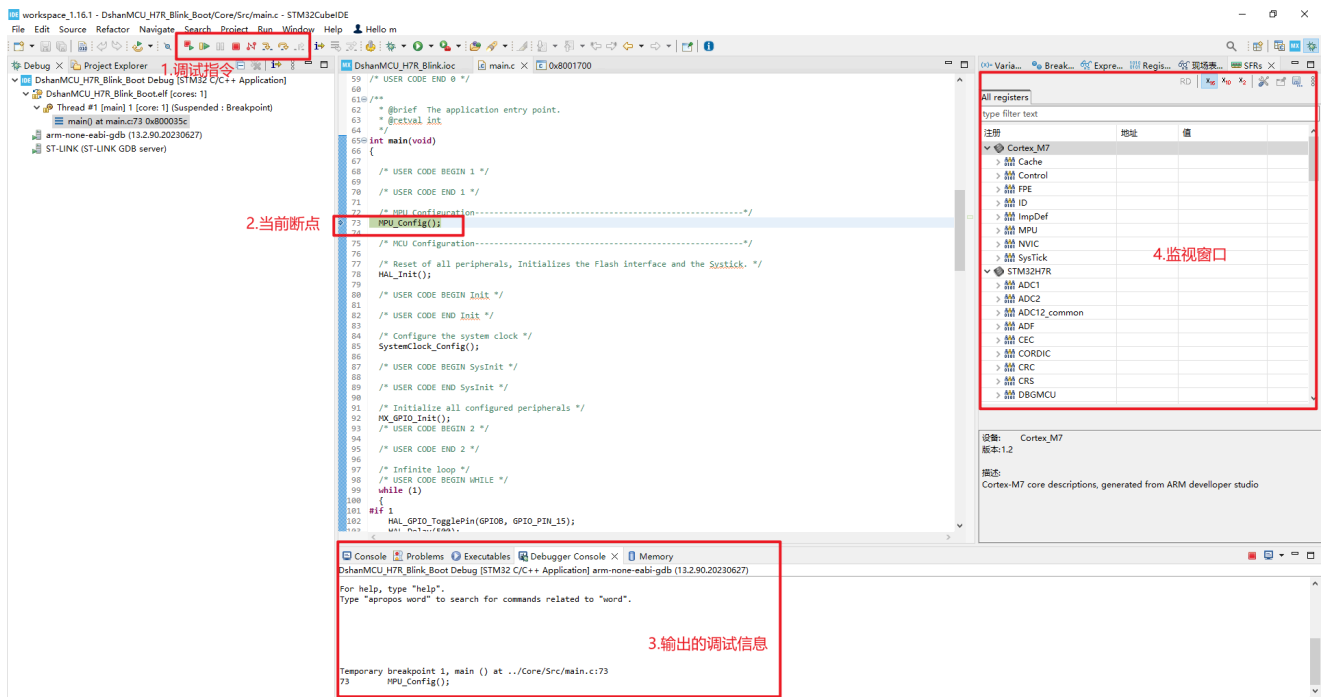
无错误，无警告，程序编译成功，无明显语法错误。

同样也可以右键点击Build Project进行编译：



烧录

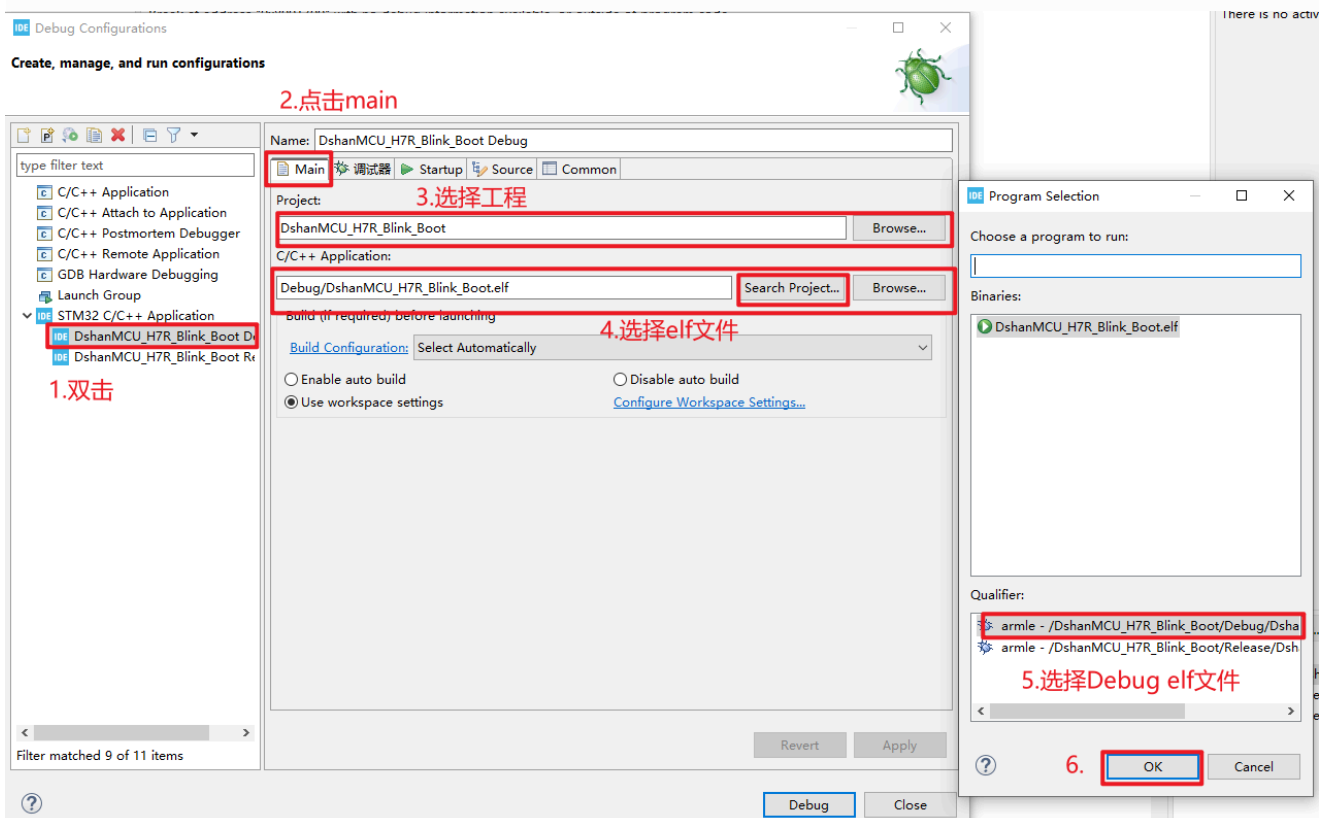
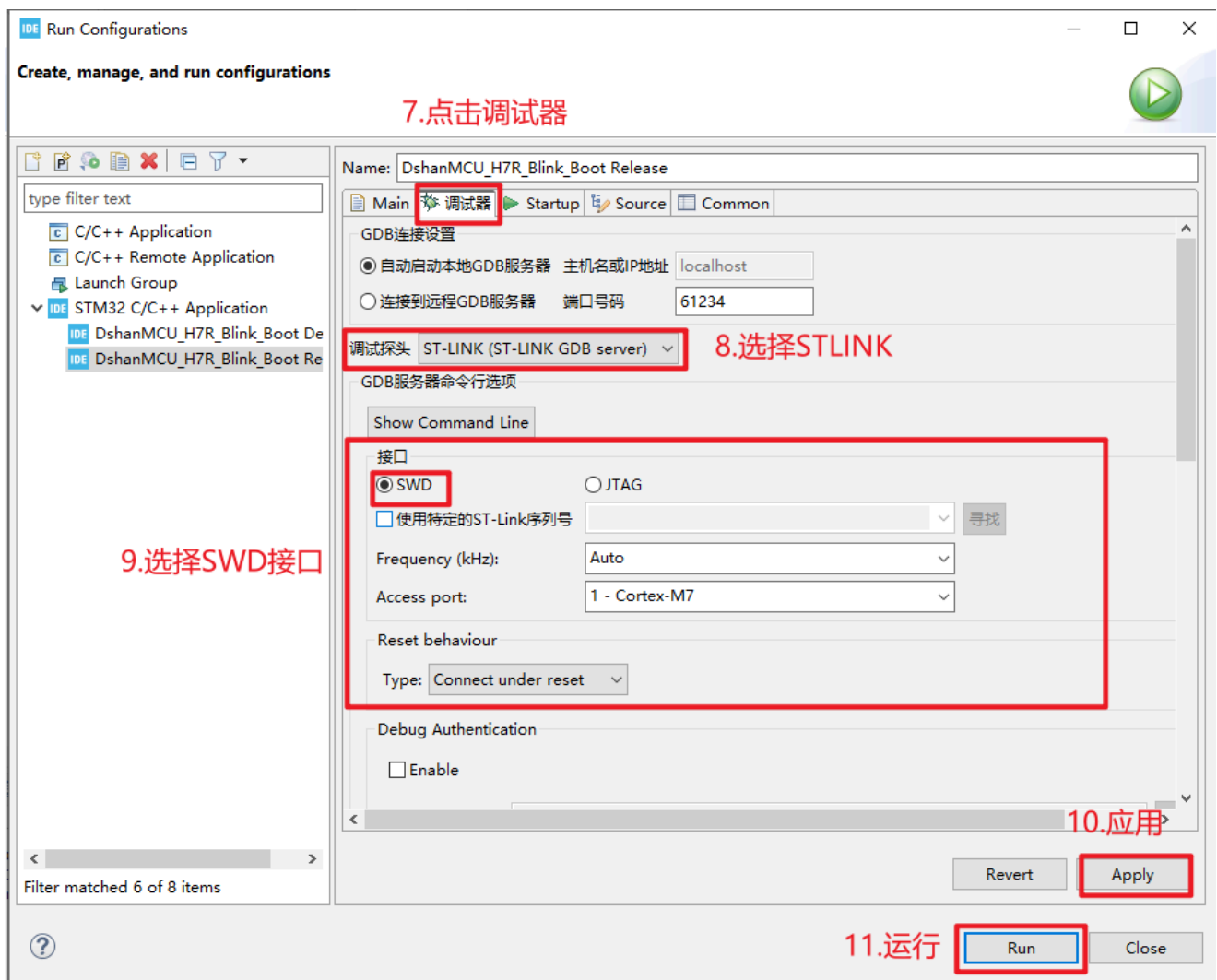
编译通过后，在工具栏Run选项中找到 **Run Configuratons**，如下图所示，配置调试用的仿真器和接口方式，最后点击 **Apply** 再点击 **Run** 即可将 *DshanMCU_H7R_Blink_Boot.elf下载到开发板中：



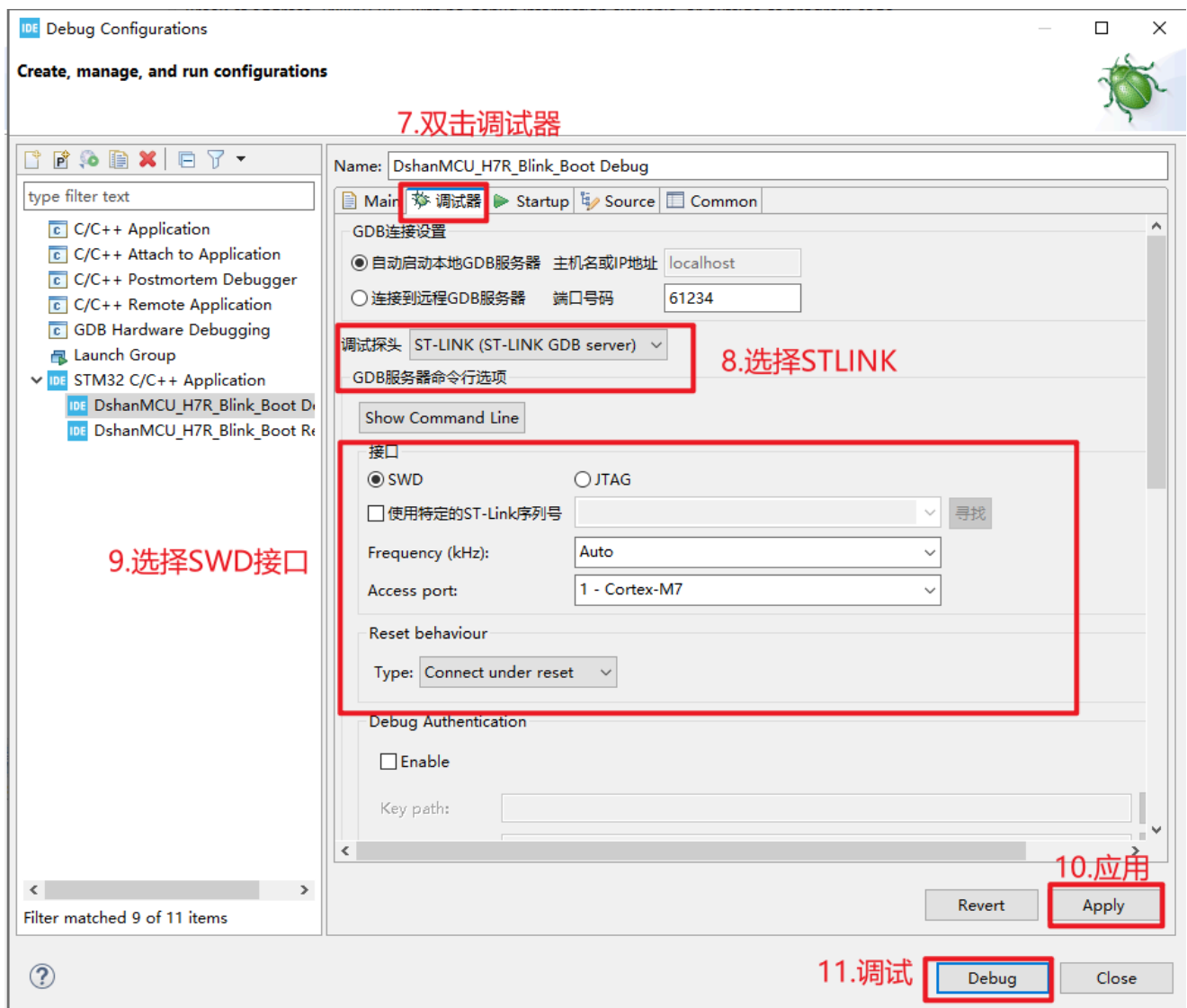
烧录完成既可以看到屏幕背光每间隔500毫秒闪烁一次。

调试程序

编译通过后，在工具栏Debug选项中找到 **Debug Configuratons**，如下图所示，配置调试用的仿真器和接口方式，最后点击 **Debug** 即可对DshanMCU_H7R_Blink_Boot.elf进行调试：



成功开始调试之后，如下图所示：



技术交流学习

欢迎加入讨论：

- 社区交流：<https://forums.100ask.net>
- QQ技术交流群（如群满，请加qq：401684796 验证备注：LVGL）：962138234
- 微信交流群：添加微信：baiwenkeji_fae 验证备注：LVGL